

Performance optimization of static mixer in urea-SCR system considering flow and turbulence characteristics

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ABSTRACT

The urea-Selective Catalytic Reduction (SCR) system is known to be the most proven technology among the various de-NO_x technologies in diesel engines. Since hydrogen and ammonia fuel are focused on every thermal system for carbon neutral policy, thermal NO_x by hydrogen, and fuel NO_x by ammonia would be increased. In the urea-SCR system, the NO_x reduction efficiency depends on the NH₃ concentration and velocity uniformity in the catalyst layer. Uniformity improves with longer flow lengths, but since the flow into the SCR is limited to 2.5 mbar back pressure, a static mixer is required to maximize uniformity while minimizing the length of the mixer to improve uniformity. The flow characteristics of blockage ratio (BR) were evaluated by the blade geometry of a static mixer used in the 2 MW class urea-SCR system. Mixing performances and flow by characteristics were investigated using numerical analysis. As a result, swirl flow was formed by the complex interaction between two blades (Blades #1 and #2) due to the geometry of the static mixer. Blade #1 increased uniformity index(UI) by 8.8 % compared to Blade #2, and Blade #2 increased turbulent intensity(TI) by 28.5 % compared to Blade #1. The influence of blade offset was analyzed. As blade offset increased, back pressure decreased by 37.5 % in all BRs, and the reactor's inlet UI_{vel} decreased by 5.9 %, 5.6 %, and 3.5 % in BR50, BR45, and BR40, respectively. The cases satisfying all constraints with the maximum mixing performance and back pressure were "BR40-1-0.9D" or "BR40-1-1.2D".

1. Introduction

Nitrogen oxides (NO_x) emissions generally referred to as nitric oxide (NO) and nitrogen dioxide (NO₂) are generated by combustion processes in industries, such as power generation boilers and internal combustion engines. It is known that NO_x reacts with chemicals and organic compounds in the atmosphere to directly affect the environment and human body adversely [1,2]. The International Maritime Organization (IMO) applied stricter Tier 3 to ships operating in emission control areas after January 1, 2016 [3]. As shown in Table 1, the Tier 3 of NO_x emissions regulation stipulated by IMO must satisfy an additional reduction of 75 % compared with Tier 2. To satisfy this strict regulation, many studies have been conducted, and various methods have been proposed.

Technologies, such as low-temperature combustion and exhaust gas recirculation, are employed to reduce NO_x emissions. However, because such technologies reduce NO_x emissions in the combustion stage, they have the disadvantage of having to change the existing engine and combustion mechanism [4]. According to the carbon neutral policy, hydrogen and ammonia fuel should be used, and they generate lots of thermal and fuel NO_x. Therefore, it is necessary to reduce NO_x only via exhaust gas after-treatment technology, which compensates for the disadvantage. The Selective Catalytic Reduction (SCR) system is the most promising of exhaust gas after-treatment methods used to meet the strict emission regulation. In particular, the urea-SCR system, which uses urea as a reducing agent, is an efficient and suitable technology [5]. When Urea Water Solution (UWS) containing urea as a reducing agent is

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Table 1
NO_x emissions regulations stipulated by IMO.

Tier	Ship construction date on or after	Total weighted cycle emission limit (g/kWh) n = engine's rated speed (rpm)		
		$n < 130$	$n = 130\text{--}1999$	$n \geq 2000$
1	January 1, 2000	17.0	$45 \cdot n^{-0.2}$	9.8
2	January 1, 2011	14.4	$44 \cdot n^{-0.2}$	7.7
3	January 1, 2016	3.4	$9 \cdot n^{-0.2}$	2.0

sprayed on an exhaust gas manifold through which high-temperature exhaust gas flows, NH₃ is generated by evaporation and thermal de-composition [6]. Thus, the produced NH₃ reduces NO_x emissions by reducing NO_x contained in the exhaust gas to nitrogen inside the catalyst. However, the urea-SCR system has the disadvantage of NH₃ slip investigated by previous studies [7–9]. In addition, the urea-SCR system has a limited distance from the outlet of the static mixer to the inlet of the catalyst due to the constraints on the installation space [10]. Therefore, a mixing device is used to satisfy high NO_x reduction performance and low NH₃ slip. In particular, static mixers are usually installed to improve the decomposition rate for UWS and increase the uniformity of spatial distribution of NH₃ and HNCO [11].

To reduce the distance required for mixing in the field of urea-SCR, many researchers have studied static mixers. First, Oh et al. [4] analyzed the factors to be considered for the design of a static mixer, stating that exhaust pipe diameter, injector type, injector performance, injector attachment angle, exhaust gas flow field, temperature, and flow rate should be considered when designing a static mixer. In addition, Zhang et al. [12] analyzed two turbulence patterns generated by a designed static mixer: swirl and turbulence flow. Further, by analyzing the mixing index and uniformity index (UI) between exhaust gas and UWS, the effect of turbulence generated by the shape of a mixer on NO_x reduction efficiency was studied [13]. Eventually, the two turbulence patterns showed a difference in mixing length, which was explained based on the dissipation rate of formed vortices. Thakur et al. [14] explained and compared static mixers of various shapes used in industrial processes to analyze the main parameters required to select suitable mixers for each system environment. As a result, it was confirmed that the evaluation of various UI had the most direct effect on the NO_x reduction efficiency of static mixers. In addition, Huang et al. [15] analyzed the mechanism of urea deposit formation due to the mixer installation and studied the optimization of mixer shape development to prevent it. Studies related to the urea-SCR system, where experimental studies are limited, have been widely conducted in the field of static mixers using computational fluid dynamics (CFD). In particular, studies on the models used for designing and optimizing mixers with injection characteristics [16] and evaluating SCR performance have been conducted [17]. In addition, previous studies have determined direct performance through NH₃ concentration UI and chemical reactions according to the static mixer shape. However, there is a lack of explanation of the mechanism of generating turbulence due to the blade shape, and existing static mixers have the disadvantage of having to bear high back pressure (BP) due to securing mixing performance.

Unlike previous studies, the following method may be used when it is difficult to directly measure NH₃ concentration UI: evaluating the performance of a static mixer and preventing NH₃ slip through NH₃ concentration UI analysis in a catalyst's front surface. The obtained NH₃ concentration UI correlates proportionally to the velocity UI in the same flow field. That is, a phenomenon in which NH₃ concentration is intense in an area with low velocity is observed [18]. Therefore, a comprehensive performance evaluation of a static mixer may be performed through velocity UI analysis. In this study, the static mixer shape of the urea-SCR system scheduled to be installed in the 2 MW class engine was considered. Assuming the proportional relationship, the change in the shape of two blades (Blade #1 and #2) with different roles in the static

mixer and the change in flow characteristics according to the installation distance were analyzed. The commercial code ANSYS FLUENT 19.0 was used for numerical analysis. Therefore, the performance of the static mixer was analyzed only by velocity UI, and the geometry optimization of the system was performed to maximize system performances, such as mixing performance and BP, through velocity UI, Turbulent Intensity (TI), and BP analyses. Therefore, a new type of static mixer that has two blades is proposed in this study.

2. The static mixer for 2 MW diesel engine and operating conditions

2.1. Static mixer

The static mixer applied to the 2 MW class diesel engine was modeled as shown in Fig. 1.

Exhaust gas from a diesel engine contains a large amount of NO_x. NO_x is removed through reactions with a catalyst in the SCR system downstream of the diesel engine. Before the exhaust gas flows into the SCR system, it is supplied after being mixed with UWS and exhaust gas. The NO_x removal reaction in the SCR system increases when the UWS and exhaust gas are mixed evenly. Therefore, the higher the uniformity of UWS and exhaust gas, the more effectively NO_x can be reduced in the SCR system. In addition, the longer the length of the mixer, the more the mixing uniformity of exhaust gas and UWS increases, but the back pressure increases, which hinders the performance in SCR. Therefore, in order to increase the mixing uniformity while having the minimum length to satisfy the back pressure required for SCR, a study on a mixer using two blades was conducted.

Fig. 2 shows the static mixer including its blades which make swirling and turbulent flows. To secure stable simulation, the length of pipe was selected as 3D and 3.6D before and after the mixer, respectively, based on the diameter of exhaust pipe. The exhaust gas flows in the y-axis direction, and the UWS is injected at the $y = 3D$ position. The length of the mixer was 2.2D. Fig. 2(b) shows the shape and installation location of the blades inside the mixer. The mixer has two blades to mix exhaust gas and urea, the concept of two blades is that make swirling flows in various direction to mix the flows. In zx plane, blade #1 and blade #2 block the flows. In yz plane, the side view of half-moon shaped blade #1 and fan shaped blade #2 are shown. In xy plane, the installed positions of blades are represented; each blades are installed in 90°, and the distance between blade #1 and #2 are research parameter. Blade #1 deflects the flows and induce tangential flow. Blade #2 collects the deflected flow and converges it to enhance the swirling flow. The exhaust flow enters to the mixer (0D position), and the urea is supplied into the upstream of the static mixer at 3D position, exhaust gas with urea collides with first blade. The turbulent flow is generated by sprayed urea and physical blockage (Blade #1). In order to optimize the static mixer that mixes urea and exhaust gas well by swirling and turbulent flow while satisfying the design back pressure, the size of blade #1 was determined as α , β , the parameter for optimizing blade #2, γ , and the parameter for optimizing the turbulent effect between the two blades, δ , were set.

The main parameters for the optimized geometry of the static mixer are α , β , γ , and δ ; other fixed dimensions are expressed as the exhaust pipe's diameter ratio. In particular, the blockage ratio (BR) is defined as the ratio of the total area to a blocked area which is excluded the red part in the static mixer's front view. As such, an increase in BR means that the area for flow is reduced. BR is expressed in Eq. (1):

$$\text{Blockage ratio (\%)} = \left(\frac{A_{\text{blocked}}}{A_{\text{total}}} \right) \times 100\% = \left(\frac{A_{\text{total}} - A_{\text{pass}}}{A_{\text{total}}} \right) \times 100\% \quad (1)$$

The analysis conditions according to BR are summarized in Table 2; the size change in Blades #1 and #2 varies at the same percentage of

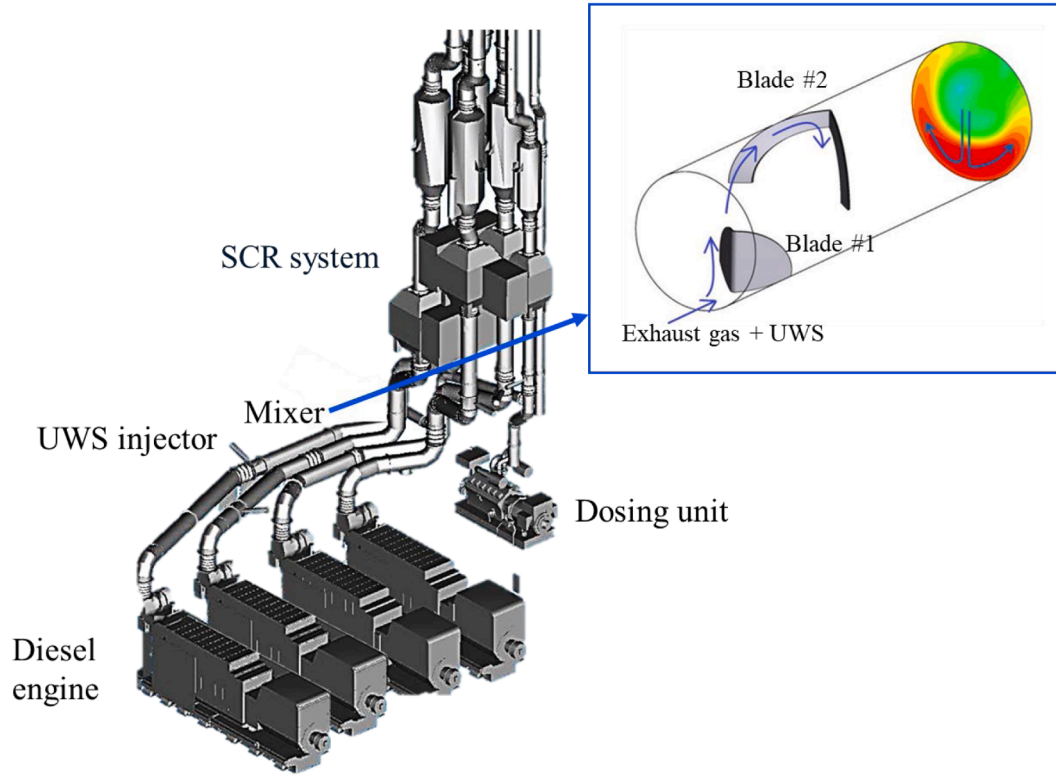


Fig. 1. The entire system of the diesel engine, mixer and SCR system.

each blade's area. For example as shown in Fig. 3., both Blades #1 and #2 in BR50 case increase 5 % more than those in BR45 case. In addition, “-1” and “-2” of cases mean that flow characteristics can be analyzed according to only the change in the size of each blade. The base model has a BR of 23.3 %, and blades #1 and #2 have identical BR ratios. The computational grid formed for CFD analysis using ANSYS Meshing (ANSYS, INC., USA) comprised about 540,000 Tetrahedrons. Skewness quality was set to ≤ 0.6 , and wall function y^+ was set to 15 [19].

2.2. Boundary conditions

The boundary conditions employed in this study were similar to the actual engine exhaust gas flow conditions used by Buckingham's Π theorem [20]. Table 3 shows the conditions that were correlated with the actual engine for each factor. Through Eq. (2) (Buckingham's Π theorem), it was confirmed that the surface roughness and Reynolds number significantly influenced the similarity. However in this study, the factor for surface roughness can be ignored because the static mixer is a smooth surface of inner-pipe with negligible roughness, which is because it is practically impossible to achieve an exact similarity of surface roughness [21]. Therefore, the boundary conditions were set by matching the Reynolds number of the actual engine exhaust pipe and the flow in the similarity model.

$$\Pi = \phi \left(\frac{l_i}{l}, \frac{h}{l}, \frac{\rho V l}{\mu} \right) \quad (2)$$

In Eq. (2), l is the characteristic length, h is the average height of surface roughness, ρ is the density of air, V is the average velocity, and μ is the viscosity of air.

3. Physical models and numerical methodology

3.1. Turbulence model

In incompressible flow, the governing equations for mass, mo-

mentum, and energy of the flow are used by three-dimensional Reynolds averaged Navier–Stoke models. Among them, the STD k - ϵ turbulence model was used for analyzing turbulence factors. The continuity, momentum, and energy equations are expressed as Eqs. (3)–(5), respectively.

$$\frac{\partial u_i}{\partial t} = 0 \quad (3)$$

$$\frac{\partial u_i}{\partial t} + \frac{\partial u_i u_j}{\partial x_j} = -\frac{\partial p}{\rho \partial x_i} + \frac{\partial}{\partial x_j} \left((v + v_t) \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right) \right) \quad (4)$$

$$\frac{\partial T}{\partial t} + \frac{\partial u_i T}{\partial x_i} = \frac{\partial}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\left(\frac{\mu}{Pr} + \frac{\mu_t}{Pr_t} \right) \frac{\partial T}{\partial x_i} \right) \quad (5)$$

The turbulent kinetic energy k and rate and rate of energy dissipation ϵ are computed from a standard two-layer k - ϵ turbulent model in Eqs. (6) and (7).

$$\frac{\partial k}{\partial t} + \frac{\partial u_i k}{\partial x_i} = \frac{\partial}{\partial x_j} \left(\left(v + \frac{v_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right) + G - \epsilon \quad (6)$$

$$\frac{\partial \epsilon}{\partial t} + \frac{\partial u_i \epsilon}{\partial x_i} = \frac{\partial}{\partial x_j} \left(\left(v + \frac{v_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right) + \frac{\epsilon}{k} (C_{\epsilon 1} G - C_{\epsilon 2} \epsilon) \quad (7)$$

G denotes the production rate of k and is given in Eq. (8).

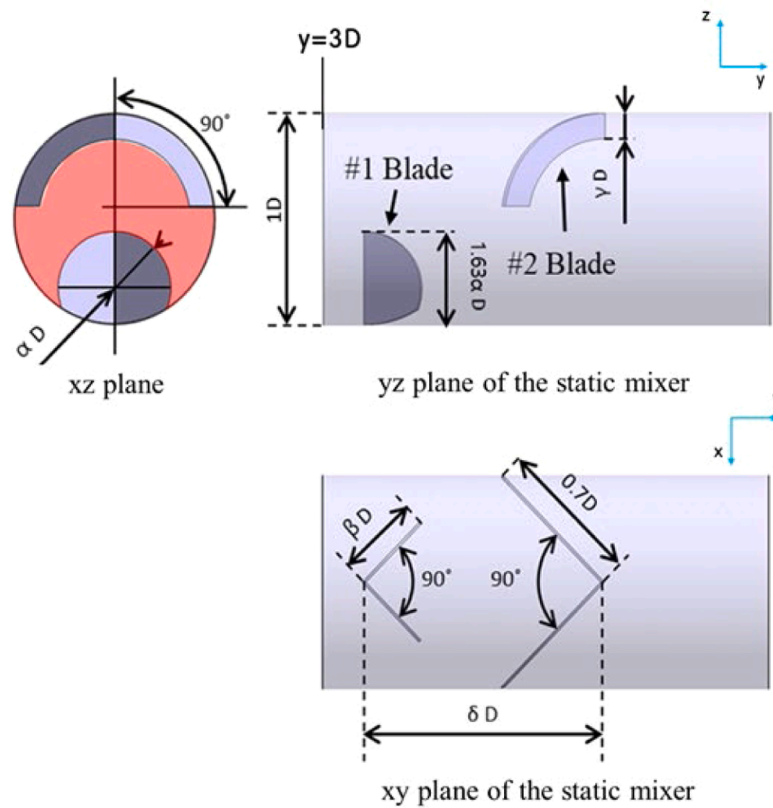
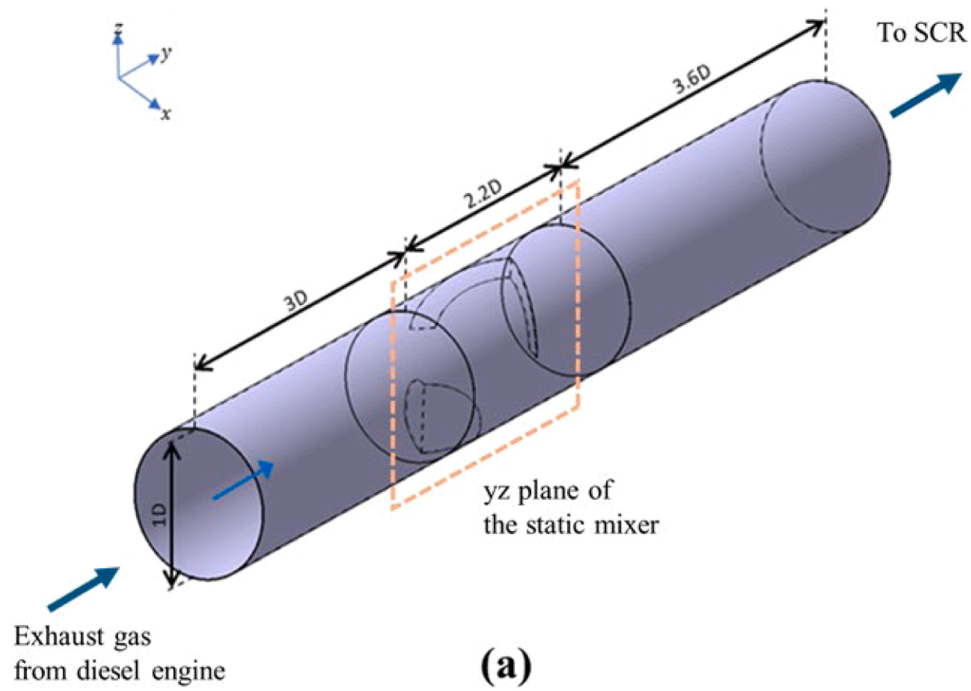
$$G = -\overline{u_i u_j} \frac{\partial u_i}{\partial x_j} = v_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \frac{\partial u_i}{\partial x_j}, \text{ and } v_t = C_\mu \frac{k^2}{\epsilon} \quad (8)$$

In the above equations, the coefficients are as follows in Eq. (9):

$$C_{\epsilon 1} = 1.44, C_{\epsilon 2} = 1.92, C_\mu = 0.09, \sigma_k = 1.0, \sigma_\epsilon = 1.3 \quad (9)$$

3.2. Uniformity index

Eq. (10) shows the UI proposed by Weltens [22].



(b)

Fig. 2. (a) Geometry of static mixer to be applied to 2 MW diesel engines and (b) cross-section view and constraints of each blade forming a static mixer.

Table 2
Computational conditions according to BR.

Cases	Blade #1 (mm)		Blade #2 (mm)
	α	β	
BR70	69.25	205.2	28.5
BR70-1	51.3	152	83.6
BR70-2	71.82	212.8	22.8
BR65	66.69	197.6	29.64
BR65-1	51.3	152	68.4
BR65-2	69.26	205.2	22.8
BR60	64.13	190	28.5
BR60-1	51.3	152	57
BR60-2	66.69	197.6	22.8
BR55	59	174.8	26.22
BR55-1	51.3	152	38.76
BR55-2	61.56	182.4	22.8
BR50	56.43	167.2	25.08
BR50-1	51.3	152	34.2
BR50-2	56.94	168.72	22.8
BR45	53.87	159.6	23.94
BR45-1	51.3	152	24.62
BR45-2	54.38	161.12	22.8
BR40	46.17	136.8	20.52
BR40-1	51.3	152	18.24
BR40-2	45.66	135.28	22.8
BR35	44.63	132.24	18.57
BR35-1	51.3	152	11.4
BR35-2	41.03	121.6	22.8
BR30	43.61	129.2	19.38
BR30-1	51.3	152	4.56
BR30-2	30.78	91.2	22.8

$$UI = 1 - \frac{1}{2n} \sum_{i=1}^n \frac{A_i \cdot |\varphi_i - \bar{\varphi}|}{A_{total} \cdot \bar{\varphi}} \quad (10)$$

Where n denotes the total number of cells, A_{total} and A_i denote the total area and area of cell i , respectively, and φ denotes the parameter of the component of interest. Therefore, as the UI approaches 1, it means that φ is uniform.

The NO_x reduction rate and NH_3 slip prevention in the urea-SCR system can be verified by evaluating velocity UI (UI_{vel}) and NH_3

concentration UI (UI_{con}) on the catalyst's front surface. If the concentration distribution of the mixed NH_3 becomes non-uniform at the catalyst's inlet, the reaction reduction rate with NO_x in the catalyst is lowered, and NH_3 slip occurs. In addition, this non-uniform distribution causes catalyst wear [23]. To mitigate this issue, a static mixer is installed in the urea-SCR system. In general, the static mixer increases the mixing performance by the turbulence generated by a specific shape between the NH_3 decomposed from UWS and the exhaust gas [12]. However, there is a correlation between UI_{vel} and UI_{con} . In general, when UI_{vel} is low, the local concentration is observed in a local area with a high velocity [18]. Therefore, the correlation between UI_{vel} and UI_{con} [18] can optimize the static mixer's performance, which is affected by blade design (i.e., BR, blade size, and offset between blades).

3.3. Grid dependency test and validation of numerical results

Before conducting the numerical simulation, grid dependency test is required to reduce errors that can be occurred in numerical calculations for the static mixer. The coarse, medium, and fine meshes were applied to the base static mixer model. Fig. 4(a) and (b) shows the numerical results for y velocity which indicates streamwise velocity, and (c) shows z velocity which indicates tangential velocity, which were analyzed at positions 0D, 0.1D, 0.2D, 0.3D and 0.4D. There were no notable differences among three mesh type in upstream region of blade #1, however the results with coarse mesh has low velocity in downstream results. It

Table 3
Summary of initial and boundary conditions.

	2 MW diesel engine exhaust gas condition	Similarity model condition (Air)
Pipe inner diameter [m]	0.5	0.19
Temperature [K]	633.15	298.15
Flow velocity [m/s]	30	21.23
Density [kg/m ³]	0.51	1.20
Mass flow rate [kg/h]	10,903.50	2609.15
Volumetric flow rate [m ³ /min]	353.48	36.12
Dynamic viscosity [m ² /s]	0.000056	0.000015
Absolute pressure [bar]		1.03
Reynolds number [–]		267,857.14

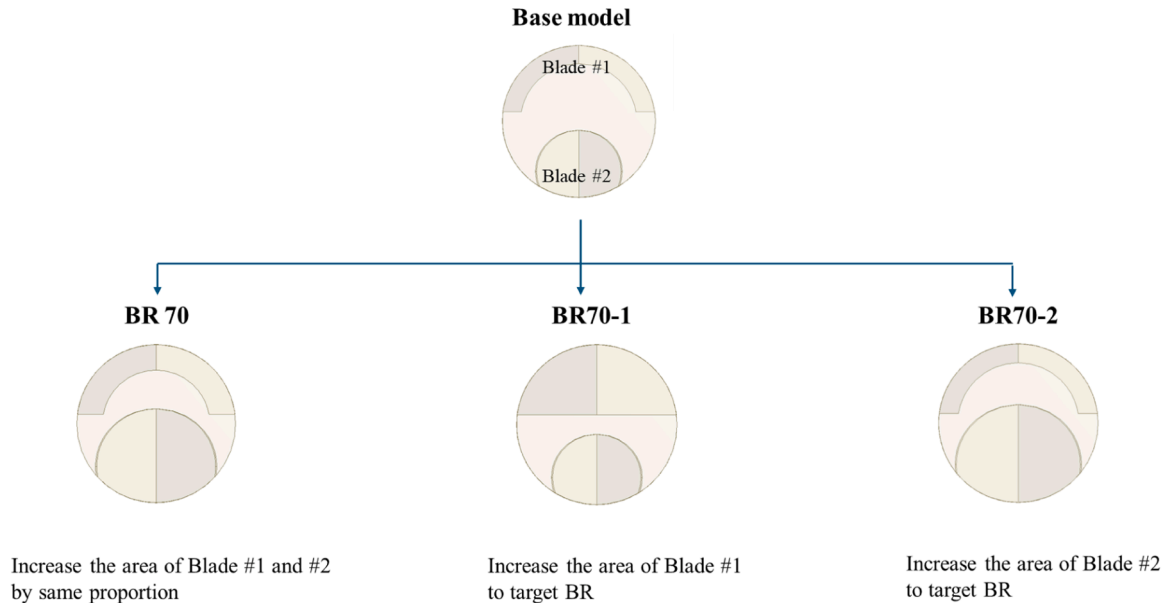


Fig. 3. Diagrammatic representation of the studied cases.

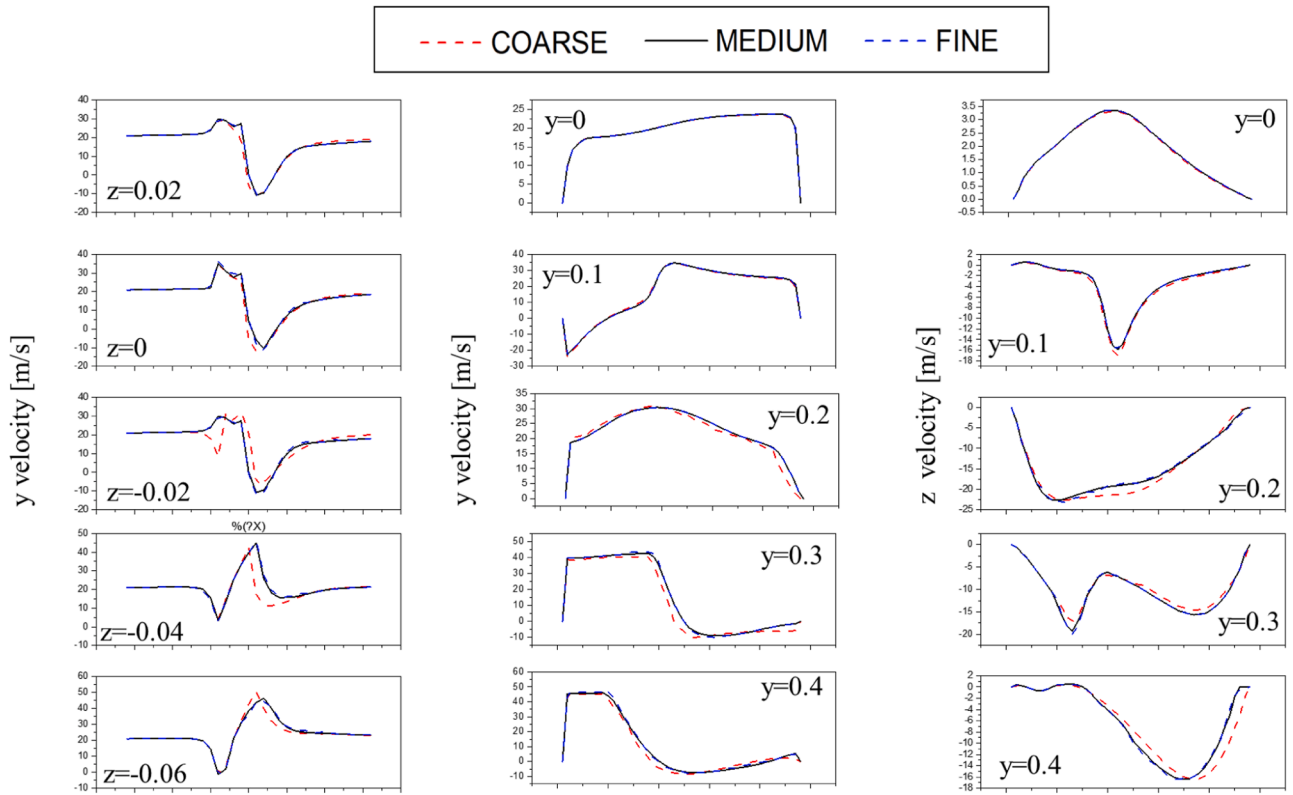


Fig. 4. Profiles of (a) y velocity for z-position, (b) y velocity for y-position, and (c) z velocity for y-positions for three different meshes.

means the coarse mesh cannot precisely calculate actual flow, vortex and swirling flows. The medium and fine meshes are similar results in all positions, therefore the medium mesh was selected considering the

computational cost and complex swirling flow.

The results of selected medium mesh were compared to experimental results to validate the numerical simulation in Fig 5. Fig. 5(a) shows the

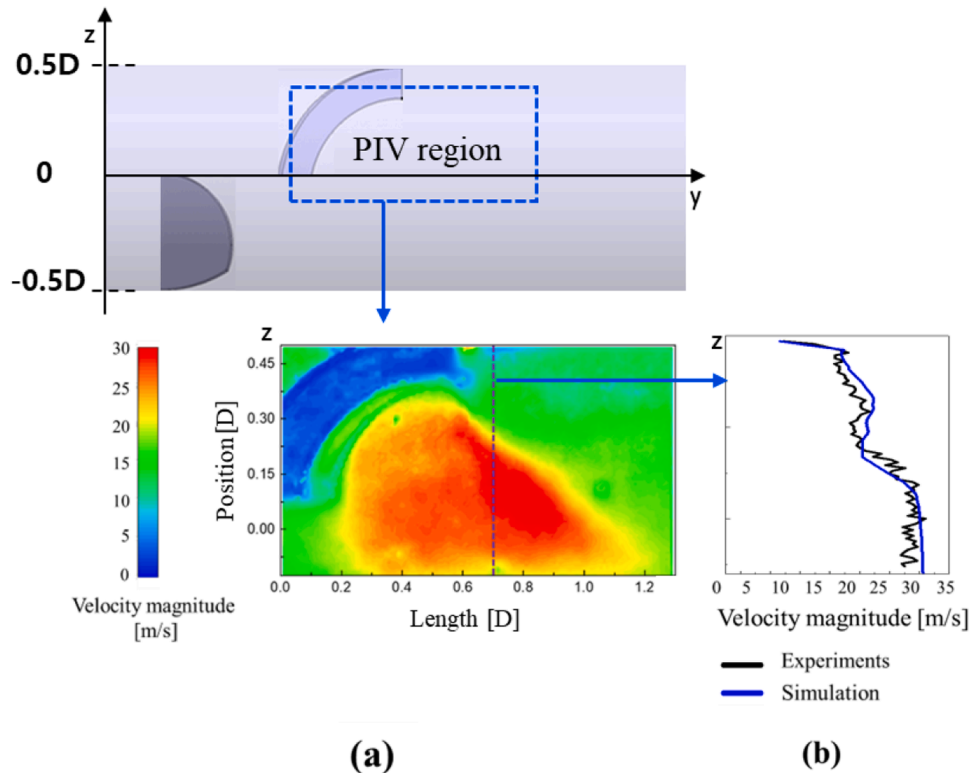


Fig. 5. (a) PIV region and experimental results, (b) comparison between experiments and simulation results at 0.7D in PIV region.

region which conducted experiments, and contour of particle image velocimetry (PIV) results. The PIV system (TSI Inc.) consisted of a double-pulse Nd:YAG laser (EverGreen 200, Quantel), which had a maximum energy output of 200 mJ/pulse at a wavelength of 532 nm, with laser sheet optics; a digital charge-coupled device (CCD) camera (TSI PowerView Plus 2MP) with a resolution of 1600×1200 pixels; a synchronizer (LaserPulse, TSI Model 610,036); and a computer. The time delay between the two PIV frames was $100 \mu\text{s}$, with a repetition rate of 14.5 Hz. The CCD camera was equipped with a Nikon 50-mm lens. If the experiment is performed in the full velocity field, the distance traveled by the flow per pixel of the image becomes too large to ensure the accuracy of the experimental results. Therefore, the experiments were performed in a certain area where turbulent and swirl flow mainly occurs designated downstream of blade 2. The magnification of the imaging lens resulted in a spatial resolution of approximately $170 \mu\text{m}/\text{pixel}$, corresponding to $272 \times 204 \text{ mm}^2$, and 500 image pairs were recorded for each case. The results were averaged for 500 image pairs at a cross-section of the static mixer using the TSI Insight TM 4 G software. The maximum velocity is about 60 m/s, and the minimum velocity is 0 m/s by structure. The fast flow occurred in the behind the structure, and the flow variation is noticeable in 0.7D position. Fig. 5(b) shows the PIV experimental and simulation results in 0.7D, the trend of velocity magnitude is very similar, and the maximum error is 9.6 %.

4. Results and discussion

4.1. Flow characteristics according to BR

To analyze the flow characteristics according to the main geometry parameters (α , β , and γ) of the static mixer, the flow pattern was first analyzed. Through flow pattern analysis, the mechanism of turbulence generated by the static mixer geometry was analyzed. Fig. 6 shows streamline pattern according to BR of Eq. (1) which is defined as the

ratio of the total area to a blocked area. Blade #1 has V-shape in the flow direction. Owing to this shape characteristic, it deflects the flow to the top of the pipe and disperses it in both directions. In addition, the cross-sectional area decreases due to Blade #1, and the flow rate increases rapidly. Blade #2 has V-shape in the opposite direction of flow, and unlike Blade #1, the center of Blade #2 is empty, as shown in Figs. 1 and 2. Therefore, the previously deflected and distributed flow is concentrated to the center of the pipe, and rotation momentum occurs. Because of these effects, the downstream of Blade #2 forms a swirl flow of clockwise and counter-clockwise at the center of the pipe axis direction. It can be observed that this swirl flow becomes stronger as BR increases. As BR increases, the area through which the fluid can flow decreases for constant flow rate, so the velocity between the blades increases. This increase in velocity strongly forms a vortex downstream of blade 2.

Fig. 7 is a diagram showing a vector for a cross-section at a specific location downstream of Blade #2. Fig. 7(a) is a cross-sectional vector in the case of BR70, whereas Fig. 7(b) is a cross-sectional vector in the case of BR30. As mentioned above, the change in swirl flow according to BR was clear, and it can be seen that the swirl flow intensity increases with BR.

Swirl flow generated by various shapes of the static mixer might strengthen the mixing performance between exhaust gas and NH_3 decomposed from the UWS. In ships, the engine bay is designed narrowly due to transportation, which is the purpose of ships. For this reason, the mixer has to be designed narrowly due to transportation, which is the purpose of ships. For this reason, the mixer has to be designed with a limited distance from the static mixer's outlet to the inlet of the reactor, so BR should be determined with a strong swirl flow satisfying the back pressure limitation in this study. TI analysis was performed to quantitatively analyze the observed turbulence. TI is expressed as the ratio of the root mean square of the turbulent velocity fluctuation (u') at a particular location over a specified period to the average of the velocity (U) at the same location over the same period,

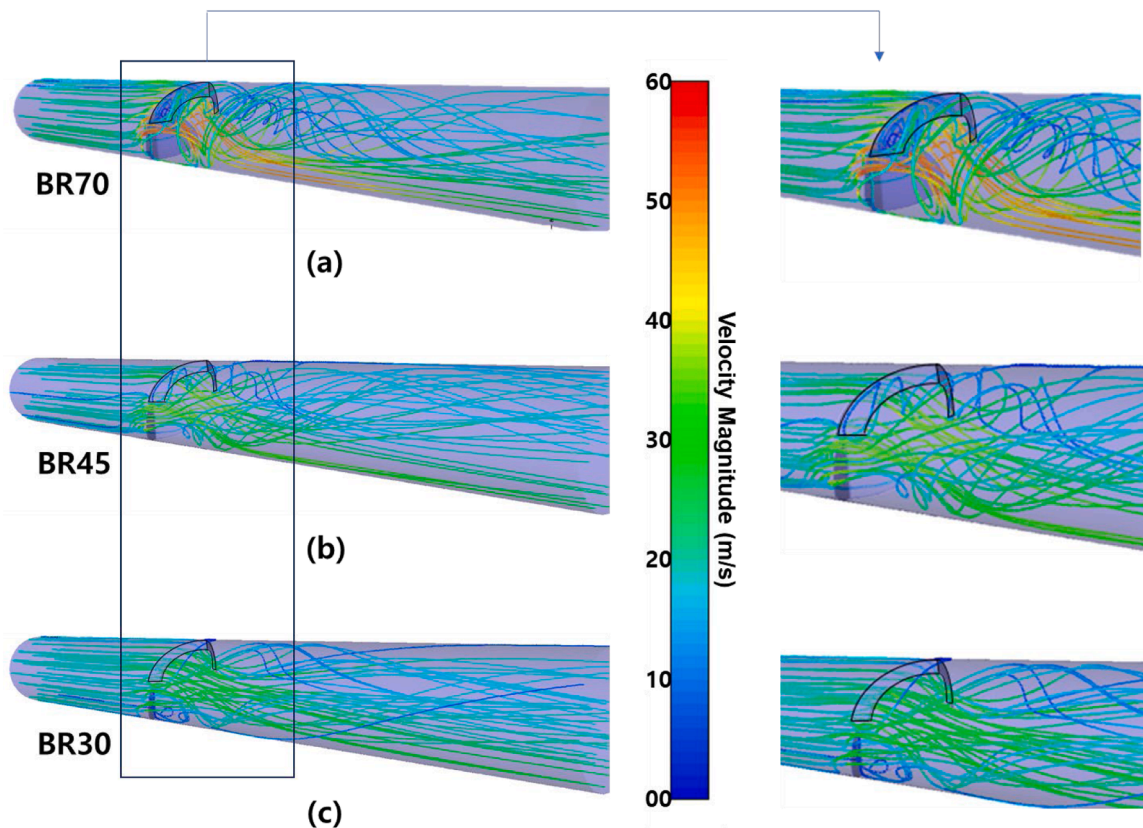


Fig. 6. Streamline pattern according to BR: (a) BR70, (b) BR45, and (c) BR30.

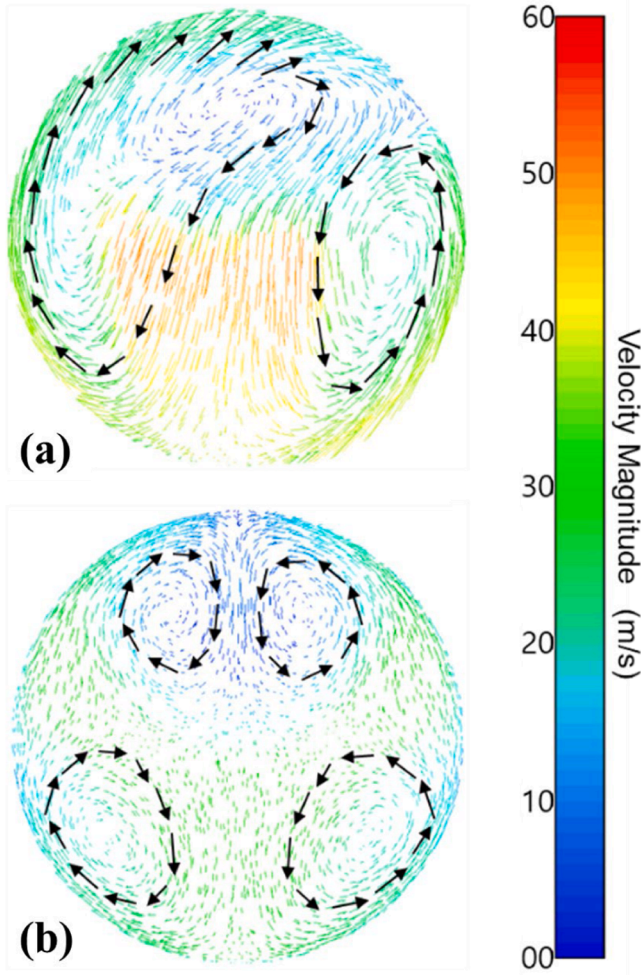


Fig. 7. Vector at the same location downstream of the static mixer: (a) BR70 and (b) BR30).

and the unit of the value is percent (%). The expression is expressed in Eq. (11):

$$\text{Turbulent intensity (\%)} = \frac{u'}{U} \times 100\% \quad (11)$$

Fig. 8 shows the contour of TI according to BR. First, the turbulence that occurred downstream of Blade #1 made a local recirculation zone that occurred due to geometry characteristics of Blade #1. It was observed that as BR increased, the strength for recirculation also increased. The strong turbulence generated downstream of Blade #2 was observed by swirl flow. This is the zone after Blade #2 where direct mixing takes place. Similar to Blade #1, as BR increased, TI also increased for Blade #2. However, excessively formed turbulence caused a strong pressure drop inside the pipe. The maximum turbulent intensity increases from 763 % to 1194 % as increasing BR from 30 to 70.

Fig. 9 depicts the change in total pressure according to the position in the longitudinal direction of the static mixer. Fig. 9(a) shows a change in total pressure by various BR, and Fig. 9(b) is a graph analyzing only the size change of Blade #2 while maintaining the size of Blade #1 (“-1” cases in Table 2), and Fig. 9(c) shows total pressure analyzing only the size change of Blade #1 while maintaining the size of Blade #2 (“-2” cases in Table 2). Comparing Figs. 6 and 7(a), the largest pressure drop was observed at BR70, which had the highest BR. Conversely, BR30 showed the smallest pressure drop. When TI was strongly formed, the pressure drop increased. In addition, for BR30-1 and BR30-2, if the size of Blade #1 or #2 is extremely large or small, a pressure drop will be

relatively small. Through this, strong TI was not formed with small BR cases. Therefore, it is expected that the two blades should be designed with an appropriate BR to secure mixing performance by TI and pressure drop. Moreover, excessively high TI might deposit NH_3 at the bottom of the pipe due to the characteristics of the flow pattern. These deposits can degrade the mixing performance of the entire urea-SCR system. In addition, strong turbulence increased the pressure drop of the entire system, so the BP, which prevents a smooth discharge of exhaust gas, can reduce engine power. Therefore, each blade geometry of the static mixer should be determined considering the mixing performance and BP. Thus, various factors, such as flow characteristics and BP, were analyzed according to BR change, and the effect of each blade on the mixer's performance was analyzed.

4.2. Effect of each blade according to BR change

UI_{vel} and TI were compared to analyze the impact of each blade on static mixer performance. A cross section at the location from 0D to 8.2D in the longitudinal direction of the static mixer, the average of each factor was calculated. Velocity magnitude and UI_{vel} were first compared, as shown in Figs. 9 and 10. Figs. 9(a) and 10(a) show velocity magnitude and UI_{vel} when the BR of only Blade #1 is changed, respectively. Figs. 9(b) and 10(b) show velocity magnitude and UI_{vel} when the BR of only Blade #2 is changed, respectively.

As shown in Fig. 10, the maximum difference in velocity magnitude was observed to be 51.90 % when the geometry of Blade #1 was changed (BR70-2, BR30-2). The maximum difference in velocity magnitude of Blade #1 was higher than that of Blade #2. In addition, the maximum difference of UI_{vel} was observed to be 43.27 % when the geometry of Blade #1 was changed (BR70-2, BR30-2), which was higher than the result of Blade #2. Based on these results, it was judged that the role of Blade #1 caused a change in the flow pattern. The reason the change in velocity magnitude and UI_{vel} increased with the size of Blade #1 was that the BR was dominantly affected by the size change of Blade #1 and the increased BR by Blade #1 formed a rapid velocity gradient. As mentioned in Section 4.1, the swirl flow could not be formed with an extremely small size of Blade #1. Therefore, Blade #1 reduced UI_{vel} in the preparation process, meaning that the blade disturbed the uniformity of swirl flow by deflecting and dispersing the UWS sprayed to the top of the pipe and exhaust gas.

Moreover, in Fig. 11(b), the graphs BR70-1, BR65-1, and BR60-1 show a temporary decrease of >10 % at 4D-5D. This phenomenon occurred when the size of Blade #2 was extremely large, and the flow was excessively concentrated in the bottom of the pipe (Fig. 11), which shows the velocity magnitude contour for each BR.

Fig. 12(a)-(c) show the case of BR70-1, BR65-1, and BR60-1, respectively, which represent the case where the size of Blade #2 is extremely large. Owing to the effect of the size of Blade #2, the flow was excessively concentrated in the bottom part of the pipe than in other BR, and the flow with a velocity close to 0 m/s occupied more than half of the flow field in the upper part of the pipe. As such, a temporary decrease in UI_{vel} was observed at pipe positions 4D-5.5D.

Fig. 13 shows the change in TI according to the static mixer position. Fig. 13(a) shows them change in TI when the BR of only Blade #1 is changed, whereas Fig. 13(b) shows the change in TI when the BR of only Blade #2 is changed.

When the flow contacted Blade #1, the TI increased sharply, and the intensity was further strengthened downstream of Blade #2. When only the size of Blade #1 is increased (Fig. 13(a)), it shows a different growth pattern than when only the size of Blade #2 is increased (Fig. 13(b)). As only the size of Blade #1 changed, TI in the downstream of Blade #2 increased linearly with BR. When Blade #2 was changed, TI increased as BR increased downstream of Blade #2 (Fig. 13(b-2)). However, Blade #1 maintained its size, but TI decreased as BR increased (Fig. 13(b-1)). The reason is that when the size of Blade #2 only changed there is a pressure valley at the 4D-position in Fig. 9(c). In addition, comparing the

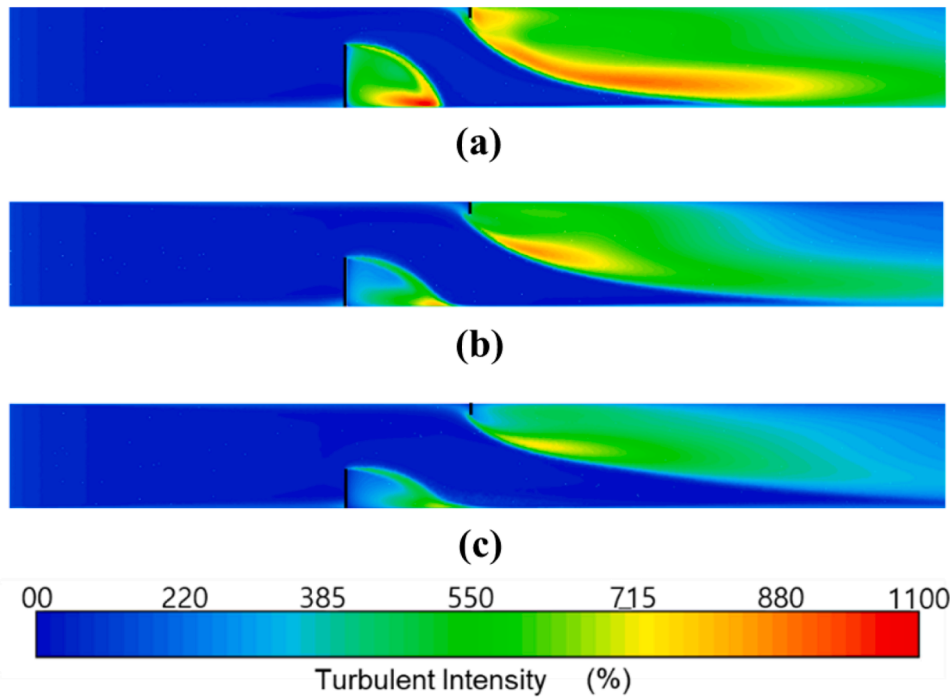


Fig. 8. TI (%) contour according to BR: (a) BR70, (b) BR45, and (c) BR30.

single blade size change under the same BR, the maximum TI was more sensitive to the size change of Blade #2 than that of Blade #1 (BR70–1: 913 %, BR70–2: 685 %). The change in TI according to the size change of Blade #1 was observed to be smaller than that of Blade #2.

Accordingly, Blade #2 concentrates the flow that has been deflected and distributed by Blade #1 to the pipe's center. In addition, Blade #2 forms a rotation momentum to generate a swirl flow and directly improves the mixing performance of the static mixer. So, Blade #2 generates swirl flow for enhancing mixing performance in the static mixer. It was possible to form an effective flow pattern for mixing through the geometry of a static mixer designed in two stages, including Blades #1 and #2. The effect of each blade causing effective flow characteristics for mixing was analyzed on the performance of the static mixer. Afterward, the geometry optimization of the blade was performed by comparing the correlation between mixing performance and BP.

4.3. The correlation between UI_{vel} and BP, and effect of blade offset

The exact dimensions of the two blades are determined through optimization between mixing performance and BP. Mixing performance can be explained as the amount of change in the UI_{vel} according to the progress of flow by the proportional relationship between the UI_{vel} and NH_3 UI_{con} [18]. Therefore, the ΔUI_{vel} is defined as the mixed performance index of the static mixer. Accordingly, Fig. 14 shows the correlation between the ΔUI_{vel} and BP according to BR. It was observed that mixing performance and BP have a proportional relationship. That is, as BR decreased, mixing performance and BP decreased. Additional performance analysis was performed on blade offset which is the distance between Blade #1 and Blade #2 in Fig. 2. The conditions which results are <2.5 mbar of BP are BR40, BR45, and BR50 for considering engine design criteria. The analyzed conditions of blade offset are listed in Table 4.

Fig. 15(a) and (b) show BP and UI_{vel} , respectively, at the inlet of the reactor for each blade offset. The UI_{vel} at the inlet of the reactor is a factor that directly affects the lifetime of the catalyst and the NO_x reduction efficiency [23]. From Fig. 15(a), as the blade offset increased from 0.6D to 1.8D, BP decreased by about 37.5 % in all cases. The key role of the distance between the two blades is the integrated effects that

control the turbulent flow in the region between the two blades. The reason for this is that as the distance between the two blades increases, the flow pattern generated by one blade does not affect the other. So, swirl flow cannot be maintained, thus BP decreases even under the same BR condition. From Fig. 15(b), the UI_{vel} decreased with an increase in blade offset at the static mixer's outlet, i.e., the reactor's inlet. For BR45, BR40, and BR50, the reduction rate according to the change in blade offset was 5.6 %, 3.5 %, and 5.9 %, respectively. The reason for this phenomenon was that the location of Blade #2 was excessively close to the reactor, so the non-uniform flow by Blade #2 did not reach "fully-developed flow" because of the short distance. The manufacturer provided constraints of <2.5 mbar and >0.94 for BP and reactor inlet UI_{vel} , respectively. A case satisfying these conditions was selected, and the ΔUI_{vel} , a factor of mixed performance, was analyzed.

Fig. 16 shows ΔUI_{vel} for each blade offset; in particular, Fig. 15(a)–(c) show the cases of BR50, BR45, and BR40, respectively. The best case based on BP is written on each blade offset and BR. In terms of BP, a shorter blade offset causes unstable BP to the system in all BR. In terms of BP, a longer offset is better for the system; however, the system performance is not all good due to the UI_{vel} at the reactor's inlet and ΔUI_{vel} .

Table 5 shows BP, UI_{vel} at the reactor's inlet, and ΔUI_{vel} according to BR and blade offset. In terms of BP, "BR45/1.5D", "BR45/1.8D", "BR45–1/1.2D", "BR45–1/1.5D", "BR45–1/1.8D", "BR45–2/1.5D", "BR45–2/1.8D", and all of BR 40 % (BR40, BR40–1, and BR40–2), except for 0.6D cases, satisfied BP conditions. In addition, in terms of UI_{vel} at the reactor's inlet, "BR45–1/1.2D" and all of BR 40 % (BR40, BR40–1, and BR40–2), except for the entire "/0.6D", "BR40–1/1.5D", and "BR40–1/1.8D" cases, satisfied UI_{vel} conditions. Considering the maximum ΔUI_{vel} in these selected cases, mixing performance with BP was optimized in "BR40–1/0.9D" or "BR40–1/1.2D", cases that satisfied all conditions.

5. Conclusions

In this study, a static mixer optimization study was conducted among the components of the urea-SCR system of a 2 MW diesel engine for ships. Through the correlation between the UI_{vel} and UI_{con} of NH_3 , the performance of the static mixer was assessed only by analyzing the flow

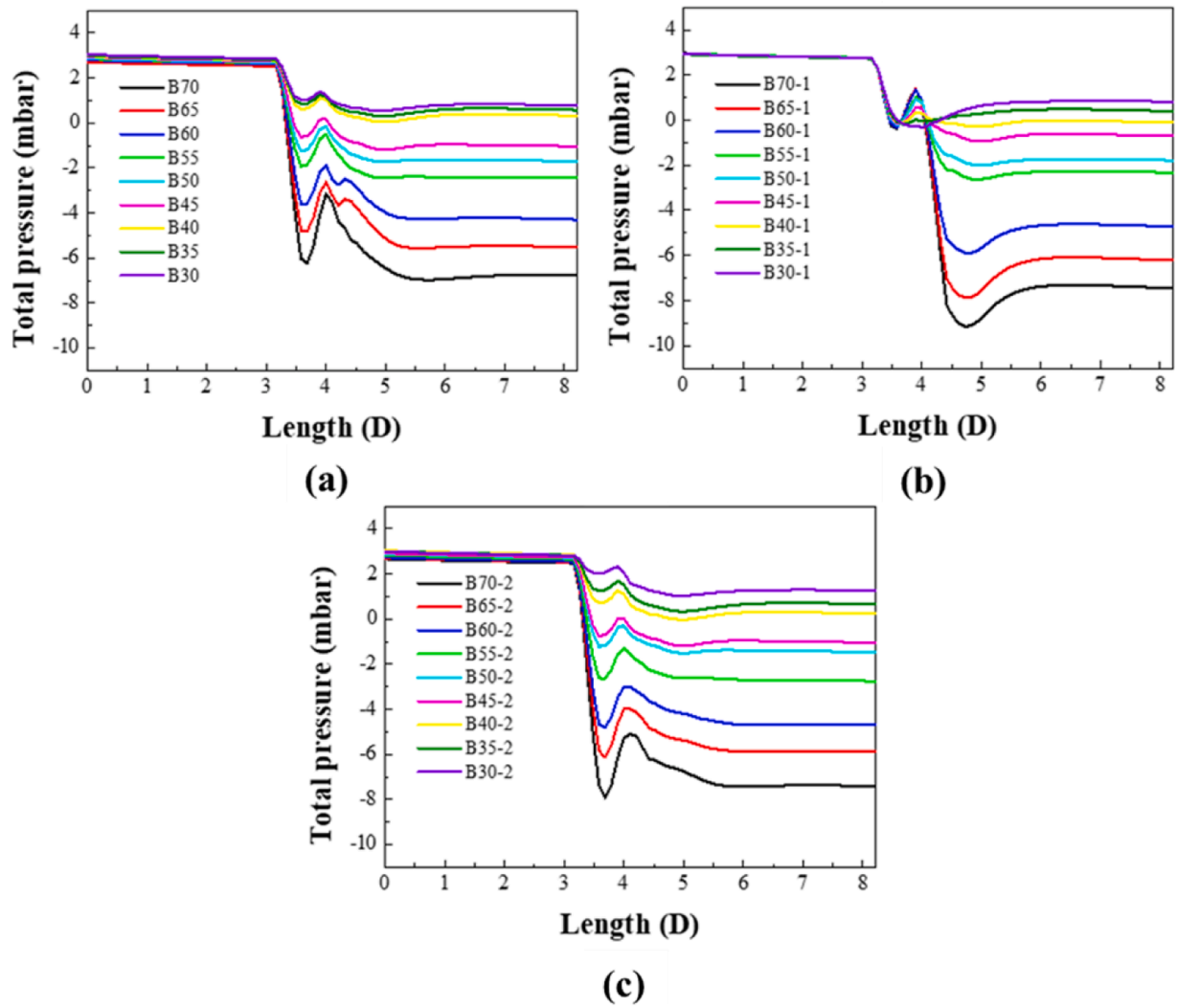


Fig. 9. Total pressure graph according to the longitudinal position of the static mixer: (a) effect of the size change of Blades #1 and #2 for the same rate, (b) effect of only the size change of Blade #2, and (c) effect of only the size change of Blade #1.

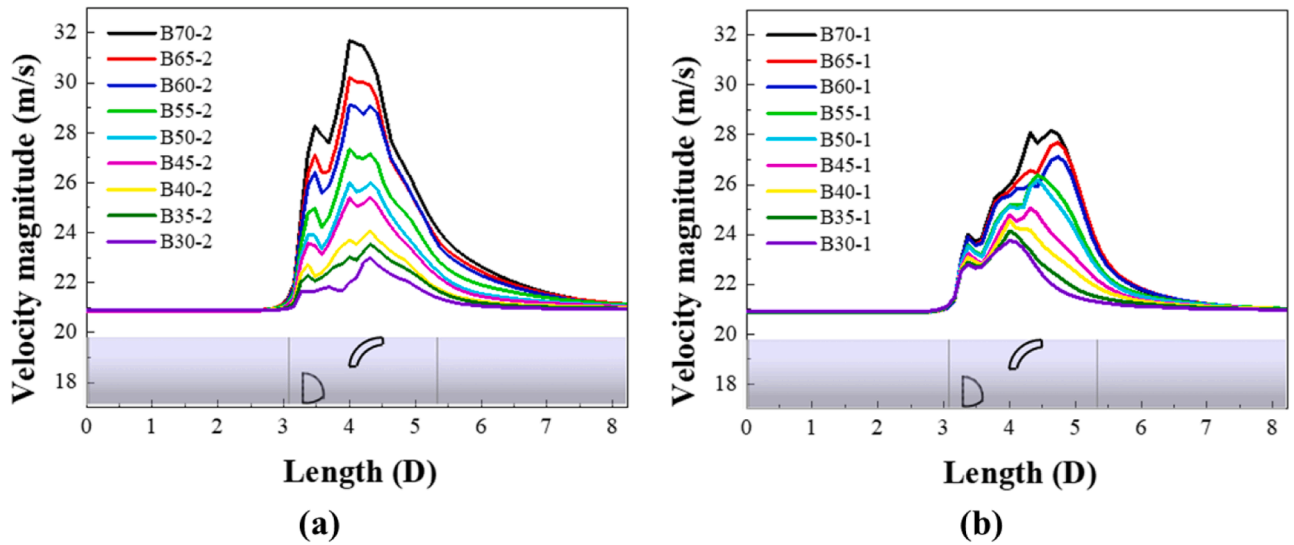


Fig. 10. Velocity magnitude graph according to the longitudinal position of the static mixer: (a) effect of only the size change of Blade #2 and (b) effect of only the size change of Blade #1.

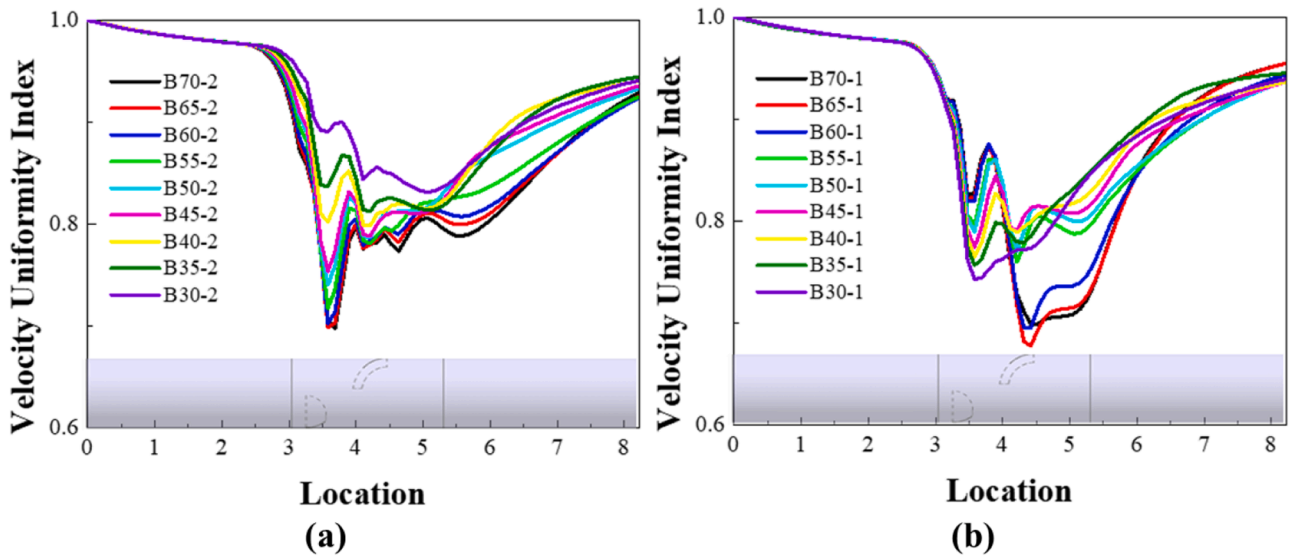


Fig. 11. UI_{vel} graph according to the longitudinal position of the static mixer: (a) effect of only the size change of Blade #2 and (b) effect of only the size change of Blade #1.

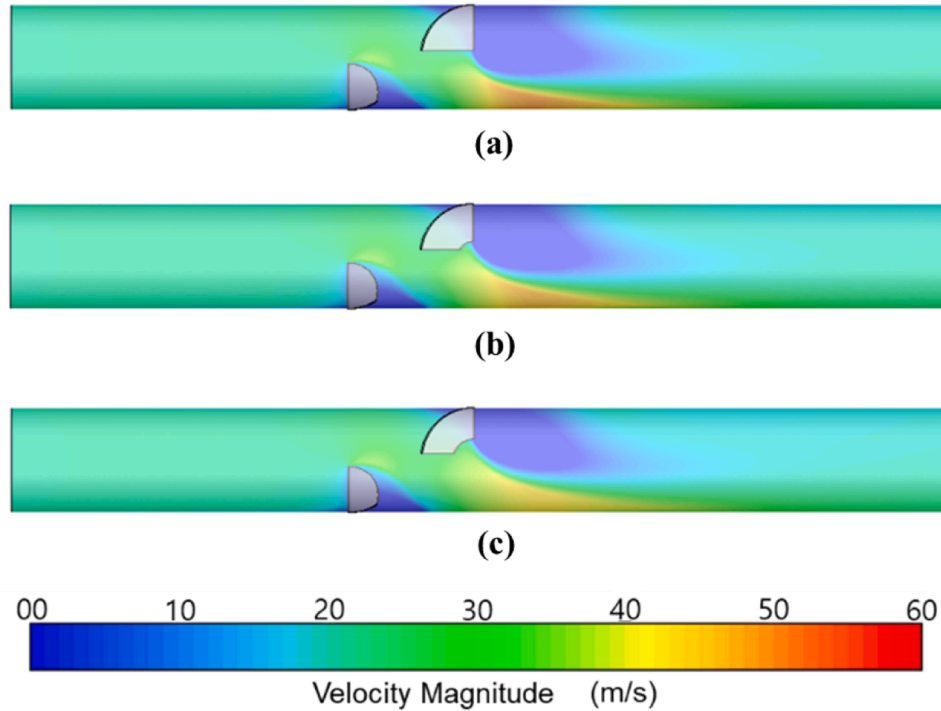


Fig. 12. Velocity magnitude contour for z-x plane according to BR: (a) BR70-1, (b) BR65-1, and (c) BR60-1.

characteristics. The static mixer comprised two blades for turbulence generation, and the blades were classified according to the BR of each blade to study the effect of blade geometry. Thus, the flow and turbulent characteristics (flow pattern, streamline, TI) and mixing performances (UI_{vel}, BP) were analyzed using CFD.

- (1) The overall streamline was observed to analyze the basic flow pattern of the static mixer. Through this, the flow deflected and dispersed by Blade #1 passed through Blade #2 and was concentrated at the center. In this case, rotation momentum was generated by Blade #2, and swirl flow was observed downstream of Blade #2. Swirl flow was formed in the clockwise and counter-clockwise directions based on the pipe longitudinal direction. It

was confirmed that the strength of this swirl flow decreased with BR.

- (2) TI was calculated for quantitative analysis of swirl flow. The strong turbulence observed from the rear of Blade #1 was due to recirculation. In addition, the strong turbulence observed downstream of Blade #2 was formed by swirl flow, the turbulent intensity increases from 763 % to 1194 % as BR increases from 30 to 70. The effect of blade #1 is higher than that of blade #2. The maximum TI by blade #1 is 930 % and that by blade #2 is 700 %, therefore the effective tool to make strong turbulent flow is blade #1 which makes tangential flow.
- (3) Change of uniformity index and back pressure decreases as decreasing blockage ratio. The conditions which satisfy the 2.5

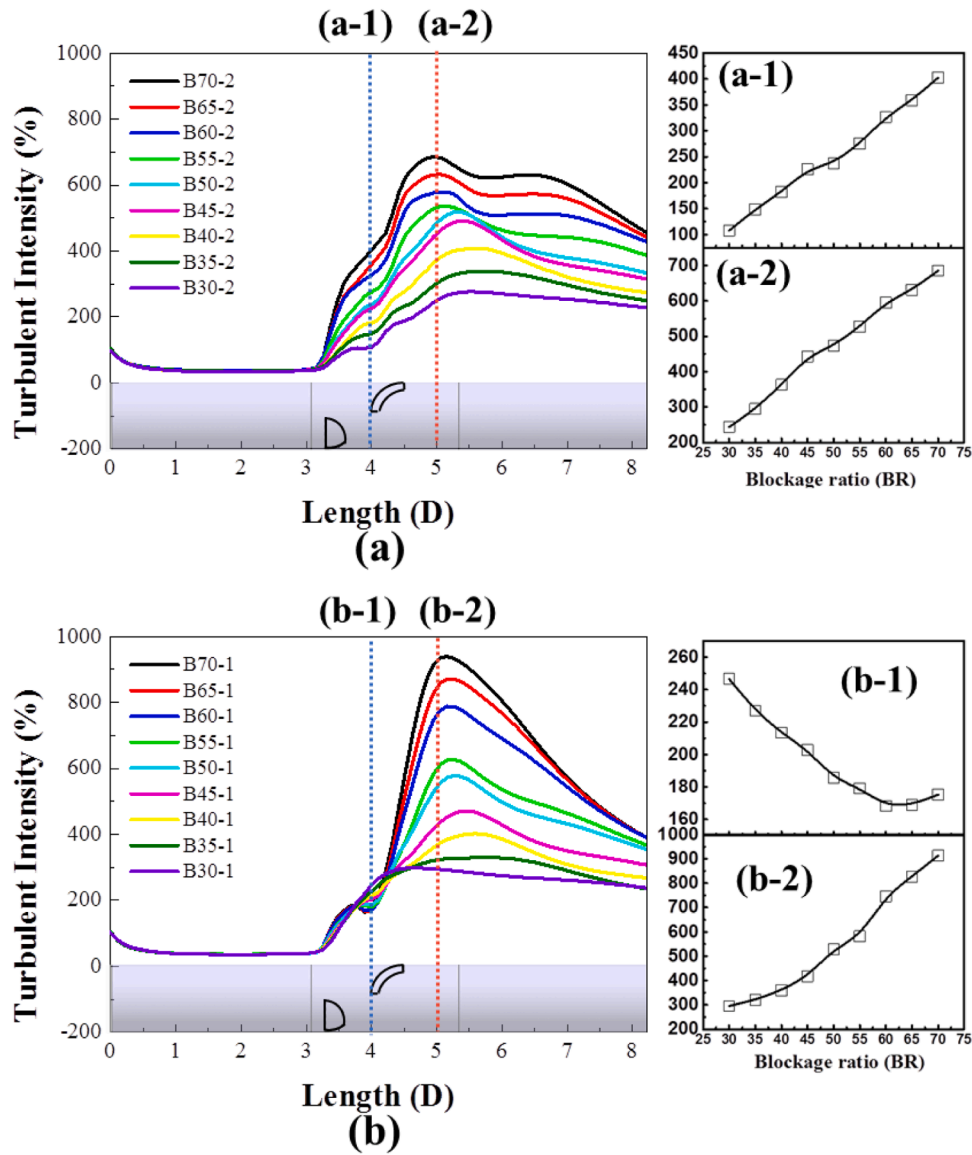


Fig. 13. TI graph according to the longitudinal position of the static mixer: (a) effect of only the size change of Blade #2 and (b) effect of only the size change of Blade #1.

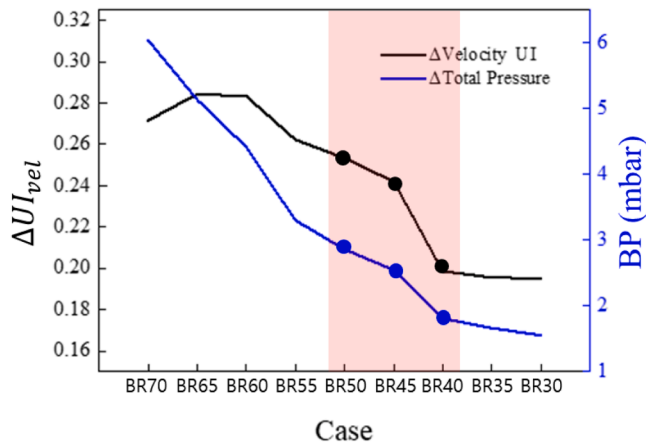


Fig. 14. Correlation graph between ΔUI_{vel} and BP according to BR.

Table 4

Computational conditions according to the blade offset.

Blade offset, δ (D)				
A	B	C	D	E
0.6	0.9	1.2	1.5	1.8

mbar of back pressure limitation are BR40, BR45, and BR50. The change of uniformity index is 0.2, 0.245 and 0.256, respectively.

- (4) For further optimization, the effect of blade offset was analyzed. When the blade offset increased, the flow pattern formed by each blade did not affect each other but adversely affected the entire system due to insufficiently developed flow to the reactor's inlet. In other words, the case showing the maximum mixing performance ΔUI_{vel} under the conditions of BP < 2.5 mbar and reactor's inlet $UI_{vel} > 0.94$ was attributed "BR40-1/0.9D" and "BR40-1/1.2D", respectively.

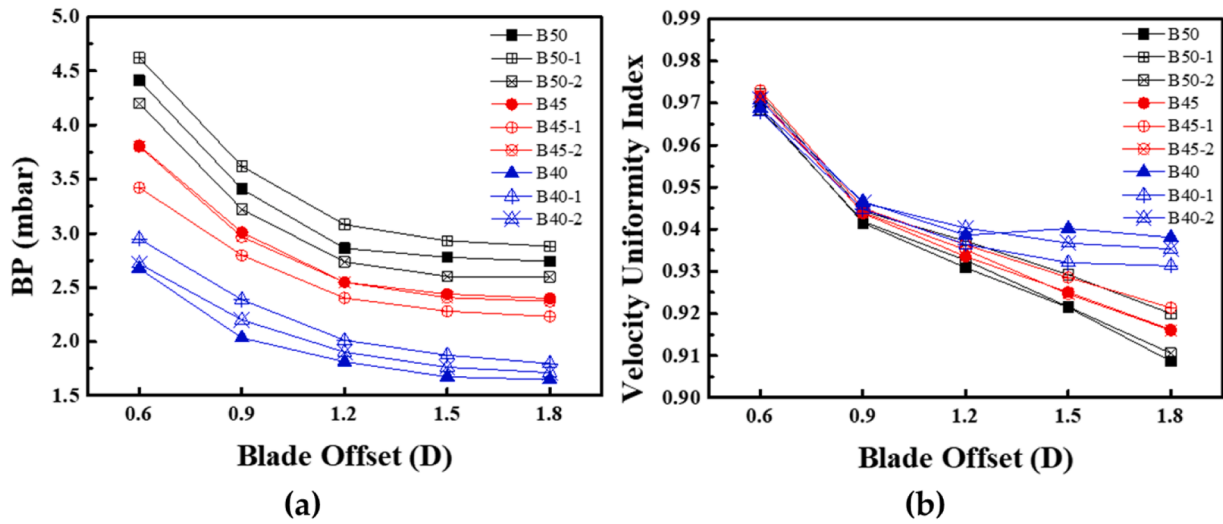


Fig. 15. BP and UI_{vel} at the static mixer outlet according to the blade offset: (a) BP for each blade offset and BR and (b) UI_{vel} at the reactor's inlet for each blade offset and BR.

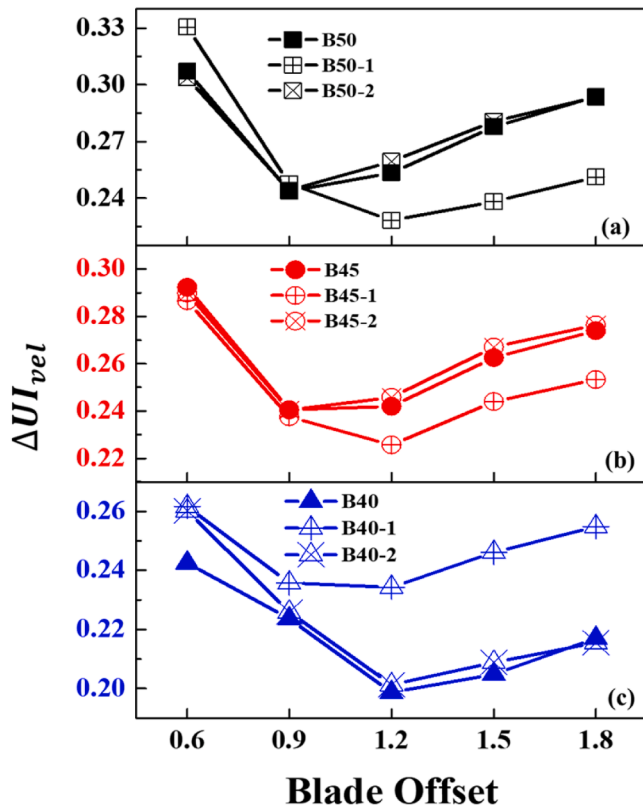


Fig. 16. ΔUI_{vel} according to blade offset and BR: (a) BR 50 %, (b) BR 45 %, and (c) BR 40 %.

CRediT authorship contribution statement

Chanho Jung: Writing – original draft, Validation, Data curation, Conceptualization. **Sanghyeon Lee:** Writing – original draft, Software, Formal analysis. **Jaehwan Jang:** Validation, Methodology. **Sangkyung Na:** Methodology, Investigation. **Gyungmin Choi:** Resources, Project administration, Funding acquisition, Formal analysis. **Minsung Choi:** Writing – review & editing, Visualization, Supervision. **Yeseul Park:** Writing – review & editing, Supervision, Funding acquisition, Data curation, Conceptualization.

Table 5

BP, UI_{vel} at the reactor's inlet, and ΔUI_{vel} according to BR and blade offset.

Blade offset [D]		0.6	0.9	1.2	1.5	1.8
BR50	BP [mbar]	4.41	3.41	2.86	2.78	2.74
	UI_{vel}	0.97	0.94	0.93	0.92	0.91
	ΔUI_{vel}	0.31	0.24	0.25	0.28	0.29
BR50-1	BP [mbar]	4.62	3.62	3.08	2.93	2.88
	UI_{vel}	0.97	0.94	0.94	0.93	0.92
	ΔUI_{vel}	0.33	0.25	0.23	0.24	0.25
BR50-2	BP [mbar]	4.2	3.22	2.74	2.6	2.6
	UI_{vel}	0.97	0.94	0.93	0.92	0.91
	ΔUI_{vel}	0.30	0.24	0.26	0.28	0.29
BR45	BP [mbar]	3.81	3.01	2.55	2.44	2.4
	UI_{vel}	0.97	0.94	0.93	0.93	0.92
	ΔUI_{vel}	0.29	0.24	0.24	0.26	0.27
BR45-1	BP [mbar]	3.42	2.8	2.4	2.28	2.23
	UI_{vel}	0.97	0.95	0.94	0.93	0.92
	ΔUI_{vel}	0.29	0.24	0.23	0.24	0.25
BR45-2	BP [mbar]	3.8	2.97	2.55	2.41	2.37
	UI_{vel}	0.97	0.94	0.94	0.92	0.92
	ΔUI_{vel}	0.29	0.24	0.25	0.27	0.28
BR40	BP [mbar]	2.68	2.03	1.81	1.67	1.65
	UI_{vel}	0.97	0.95	0.94	0.94	0.94
	ΔUI_{vel}	0.24	0.22	0.20	0.20	0.22
BR40-1	BP [mbar]	2.95	2.39	2.01	1.87	1.8
	UI_{vel}	0.97	0.94	0.94	0.93	0.93
	ΔUI_{vel}	0.26	0.24	0.23	0.25	0.25
BR40-2	BP [mbar]	2.72	2.2	1.9	1.76	1.71
	UI_{vel}	0.97	0.95	0.94	0.94	0.94
	ΔUI_{vel}	0.26	0.23	0.20	0.21	0.22

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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